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Mathematical model

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The problem is: when a sunscreen lotion of SPF 30, is applied in a layer of only $5\mu\text{m}$ instead of the calibration layer of $20\mu\text{m}$, how long can children stay out in the sun, when their skin would start to redden after 10 minutes without suncream?

Define the independent variable x in μm , as the depth in the suncream layer. $x = 0$ is at the top side exposed to the sun, and $x = 20\mu\text{m}$ (or $x = 5\mu\text{m}$) is on the skin side of the suncream layer.

The intensity of the UV light is $I(x)$, and $I(0) = I_{full}$, the intensity that is such that unprotected, light skinned children can tolerate 10 minutes of it midday on a summers day.

Exercise A

1/1 point (graded)

Define Δx as the thickness of a thin layer of sunscreen $[x, x + \Delta x]$.

Define ΔI as the change in the UV light intensity in the layer $[x, x + \Delta x]$.

ΔI is defined as:

☐ $\Delta I = I(x + \Delta x) - I_{full}.$

☐ $\Delta I = I(x) - I_{full}.$

☒ $\Delta I = I(x + \Delta x) - I(x).$ ✓

☐ $\Delta I = I(x) - I(x + \Delta x).$

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You have used 1 of 1 attempt

✓ Correct (1/1 point)

Exercise B

1/1 point (graded)

The sunscreen protects the skin because it absorbs UV light. Assume that scattering and reflection of the UV light by the sunscreen lotion can be neglected.

Construct a balance equation for ΔI .

- ☐ $\Delta I = -I(x) \Delta x$
- ☐ $\Delta I = +k I(x)$
- ☒ $\Delta I = -k I(x) \Delta x$, with $k > 0$ ✓
- ☐ $\Delta I = +k I(x) \Delta x$, with $k > 0$

Answer

Correct:

The change in I is relative to the thickness of the layer, and to the intensity itself (higher intensity to begin with gives a higher change). The change should be negative as I decreases with depth.

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i Answers are displayed within the problem

Exercise C

1/1 point (graded)

Consider the balance equation.

The unit of k is

- ☐ unit of light intensity
- ☒ μm^{-1} ✓
- ☐ no unit (dimensionless)
- ☐ μm

Answer

Correct:

$[\Delta I] = [I]$, so the remaining units of the balance equation give that the product $[k][\Delta x]$ is unitless, so $[k] = [\Delta x]^{-1} = \mu\text{m}^{-1}$.

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Exercise D

1/1 point (graded)

Divide the balance equation by Δx and take the limit for $\Delta x \rightarrow 0$.

The resulting differential equation is

☐ $\Delta I = 0$

☒ $\frac{dI}{dx} = -k I(x)$ ✓

☐ $\frac{\Delta I}{\Delta x} = -k I(x)$

☐ $\frac{dI}{dx} = -k I_{full}$

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You have used 1 of 1 attempt

i Answers are displayed within the problem

In the next section, you will solve this differential equation for $I(x)$: you will *integrate* the differential equation for $I(x)$.

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